$$f_{XY}(x,y) = \begin{cases} 2\cdot 4\,x\,(2-x-y) & 0<x<1 \\ & 0<y<1 \\ 0 & \text{otherwise} \end{cases}$$

For $0<y<1$, find $f_{X|Y}(x|y)$

For $0<x<1$,

$$f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)} = \frac{f_{XY}(x,y)}{\int_{-\infty}^{\infty} f_{XY}(x,y)\,dx}$$

$$= \frac{\cancel{2\cdot 4}\,x(2-x-y)}{\int_0^1 \cancel{2\cdot 4}\,x(2-x-y)\,dx}$$

$$= \frac{x(2-x-y)}{2\left(\frac{x^2}{2}\right)_0^1 - \left(\frac{x^3}{3}\right)_0^1 - y\left(\frac{x^2}{2}\right)_0^1}$$

$$= \frac{x(2-x-y)}{2-\frac{1}{3}-\frac{y}{2}} = \frac{6x(2-x-y)}{4-3y}$$

for $x<0$ or $x>1$

$$f_{X|Y}(x|y) = 0 \text{ for } 0<y<1$$

# MGF Properties

1. $Y = aX + b$

$$\phi_Y(t) = \int_{-\infty}^{\infty} f_X(x)\, e^{t(ax+b)}\, dx \quad \because \text{LOTUS}$$

$$E\left(e^{t(aX+b)}\right)$$

$$= e^{tb} \int_{-\infty}^{\infty} f_X(x)\, e^{(at)x}\, dx$$

$$= e^{tb}\, \phi_X(t)$$

2. $Z = X + Y$ where $X$ & $Y$ are independent

$$\phi_Z(t) = E\left(e^{t(X+Y)}\right)$$

$$= E\left[e^{tX} e^{tY}\right]$$

$$= E\left[e^{tX}\right] E\left[e^{tY}\right] \quad \text{due to indep.}$$

$$= \phi_X(t)\, \phi_Y(t)$$

3. $$Z = \begin{cases} X & \text{w.p. } p \\ Y & \text{w.p. } 1-p \end{cases}$$

$$\times\; Z = pX + (1-p)Y$$

$$\phi_Z(t) = E\left(e^{tZ}\right) = p\,E\left(e^{tX}\right) + (1-p)\,E\left(e^{tY}\right)$$

$$= p\,\phi_X(t) + (1-p)\,\phi_Y(t)$$